

Straight Line Graphs

LLE – Mathematics and Statistics

$$y = mx + c$$

1. Given the straight line function $y = 2x + 5$:
 - (a) write down the gradient
 - (b) the point where the line crosses the y -axis (the y -intercept), by substituting $x = 0$ and solving for y
 - (c) the point where the line crosses the x -axis (the x -intercept), by substituting $y = 0$ and solving for x
 - (d) make a sketch of the straight line graph
2. For each of the functions given:
 - (a) Rearrange to make y the subject
 - (b) Determine whether the function represents a straight line
 - (c) For straight lines: write down the value of the gradient, the co-ordinates of where the line crosses the y -axis, and the co-ordinates of where the line crosses the x -axis
 - (d) For straight lines: make a sketch of the function
 - i. $2y = 6x + 8$
 - ii. $y - 2x = 10$
 - iii. $xy - 5 = 10$
 - iv. $3x + 2y - 8 = 0$
 - v. $x^2 - 3y = 6$
 - vi. $x^2 - 3y^2 = 6$
 - vii. $5x - 8y = 9$

viii. $\frac{2y-x}{5} = x$

ix. $x(3y - 1) = 3xy + 4$

3. For each pair of straight lines given below:

(a) Sketch each pair on a single graph, labeling the points where the lines cross the axes (you will need to determine these).

(b) Solve them as simultaneous equations.

(c) The solution to the simultaneous equations is the point where the two lines cross. Label this point on your sketch.

i. $y = 2x + 3$

$y = x + 5$

ii. $x + y = 10$

$2x - y = 2$

4. Consider the following functions for supply and demand, relating price (P) and quantity (Q):

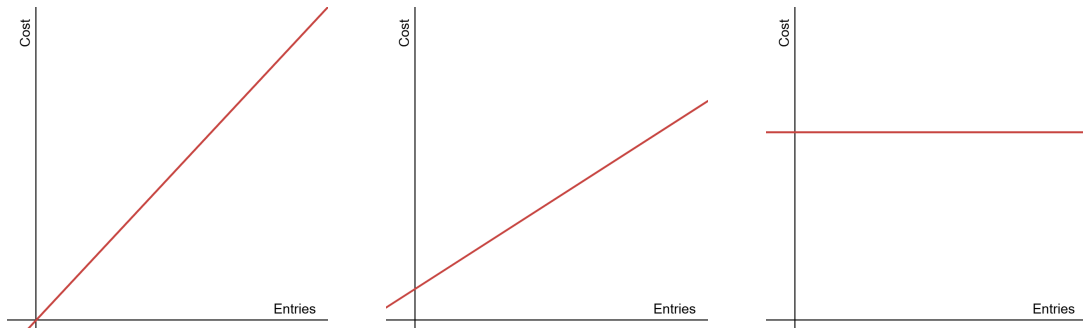
Supply: $P = Q + 18$

Demand: $3Q = 30 - P$

(a) On the same axes, sketch the supply and demand functions. Place Q on the horizontal axis and P on the vertical axis

(b) Solve the simultaneous equations for supply and demand, $P = Q + 18$ and $3Q = 30 - P$, to find the equilibrium price and quantity.

5. Consider the three graphs below which represent the total cost of a gym based on the number of entries. The point where the axes lines meet is the point $(0,0)$, no other values have been shown on the axes.



- (a) A new gym charges £20 per entry. Which of the graphs shown above best represents this pricing structure?
- (b) The manager also considers an alternative pricing structure. They offer a membership price of £20, and thereafter it only costs £12 per entry. Which of the graphs above would best represent this?"
- (c) For the unselected graph, what would be a suitable scenario for this?
- (d) Consider the first scenario, £20 per entry. Using c for cost and n for number of entries, write the formula (the equation of the line) for cost in the form $c =$.
- (e) In the £20 per entry structure, how much would it cost to enter 4 times?
- (f) Consider the second scenario, £20 fixed membership price and thereafter £12 per entry. Using c for cost and n for entries, write the formula (the equation of the line) for cost in the form $c =$.
- (g) In the £20 membership plus £12 per entry structure, how much would it cost to enter 4 times?
- (h) For both the £20 per entry (scheme 1) and the £20 membership plus £12 per entry (scheme 2), what is the maximum number of times you can attend if you have £100?
- (i) Solve the two schemes equations simultaneously, and comment on what this would mean for the customer.